

Advanced Topics in Monetary Economics with Heterogenous Agents

— Problem Set 4, Spring 2021 —

Due: Friday, March 12, 11:59 PM ET on Canvas*

Important guidelines.

1. Please only submit this problem set if you are *not auditing* this course and are *officially enrolled* for credit. Each problem set counts for 20% of your final grade.
2. Make sure to *submit your code*. The best way to solve our problem sets is using Jupyter Notebooks. We *strongly suggest using Python* from the beginning since this course will progressively build on existing Python modules in future problem sets. You can ask your classmates coding questions in Slack, Martin in OH, and review QuantEcon's programming tutorial.¹
3. The problem sets for this class are meant to give you exposure to the actual, messy research process. This means: If a question is incomplete or missing information, please go ahead and make an assumption that you deem reasonable. Be sure to *state your assumptions and submit your full derivations* in your solution. Many problems could have multiple solutions.

1 Estimating a simple HANK model

We consider an extended version of the model discussed in class that is amenable to estimation. The model features TFP shocks X_t , a government that spends an exogenous amount G_t , levies progressive taxes and follows a fiscal rule, and monetary policy that follows a Taylor rule with monetary policy shocks ϵ_t . This three-shock structure is the same as in [Herbst and Schorfheide \(2015\)](#)'s representative-agent model, which these authors use for estimation using state-space methods.

Specifically, we consider the canonical HANK model with progressive taxation at rate λ and the government adjusting the intercept of the tax schedule. Output is $Y_t = X_t N_t$, prices and flexible and wages are sticky, and the labor rationing rule is proportional so that all workers work N_t hours. As we showed in Chapter 2, this combination of assumption results in the following

*Submission of this pset is optional but encouraged.

¹<https://python-programming.quantecon.org/intro.html>

model of household decisions:

$$\begin{aligned}
V_t(a_-, e) &= \max \frac{c^{1-\sigma}}{1-\sigma} + \beta \mathbb{E} [V_{t+1}(a, e') | e] \\
c + a &= (1 + r_{t-1}) a_- + Z_t \frac{e^{1-\lambda}}{\mathbb{E}[e^{1-\lambda}]} \\
a &\geq \underline{a}
\end{aligned} \tag{1}$$

where r_t is the ex-ante real interest rate and $Z_t = Y_t - T_t$ is post-tax income. We assume that inflation is given by the following wage Phillips curve:

$$\pi_t^w = \kappa_w \left(\frac{v'(N_t) / u'(C_t)}{(1-\lambda) Z_t / N_t} - \mu \right) di + \beta \pi_{t+1}^w \tag{2}$$

where $C_t = \int c_{it} di$ is aggregate consumption and μ is the union markdown. The term $1 - \lambda$ captures progressive taxation; when $\lambda = 0$ taxation is linear and the marginal retention rate on labor is just $\frac{Z_t}{N_t} = \tau_t X_t$. Given flexible prices, $P_t = W_t / X_t$, so that inflation is

$$1 + \pi_t = (1 + \pi_{wt}) \frac{X_{t-1}}{X_t} \tag{3}$$

Monetary policy follows a Taylor rule for the nominal interest rate, so that the ex-ante real interest rate is given by

$$r_t = i + \phi_\pi \pi_t - \pi_{t+1} + \epsilon_t \tag{4}$$

where ϵ_t is a monetary policy shock, and ϕ_π is the responsiveness of nominal rates to inflation.

The government budget constraint is

$$B_t = (1 + r_{t-1}) B_{t-1} + G_t - T_t \tag{5}$$

where G_t is exogenous, and the government follows a fiscal rule for taxes,

$$T_t = T + \phi_T (B_{t-1} - B) \tag{6}$$

where ϕ_T is the responsiveness of taxes to debt. Goods market clearing implies

$$C_t + G_t = Y_t = X_t N_t \tag{7}$$

and asset market clearing implies

$$A_t = B_t \tag{8}$$

We would like to estimate this model to data on realized inflation π_t , output y_t , and nominal interest rates i_t . Since output is trending in the data, but the model is stationary, we first linearly detrend output before estimation.

1.1 Data preparation

1. Collect U.S. quarterly data on PCE inflation π_t , real output y_t , and the Federal Funds rate i_t over the period 1966Q1 to 2004Q4. Take the log of output and linearly detrend to form $\tilde{y}_t$. This delivers our data sample $(\pi_t, \tilde{y}_t, i_t)$. Report the empirical autocovariances $Cov(\tilde{y}_t, \tilde{y}_{t+k})$, $Cov(\tilde{y}_t, \pi_{t+k})$ and $Cov(\tilde{y}_t, i_{t+k})$ at various leads and lags. Comment: can you see a Phillips curve relationship in the data? What episodes are driving the correlation between output and inflation? Output and interest rates?

We now want to know what combination of shocks and parameters achieves the best fit of our model to these observed patterns.

1.2 Representative agent model

To start, let's estimate the representative-agent version of this model, similar to what [Herbst and Schorfheide \(2015\)](#) do. We replace (1) with the Euler equation

$$C_t^{-\sigma} = \beta(1 + r_t) C_{t+1}^{-\sigma} \quad (9)$$

2. Set up this representative-agent model in the sequence space. What are your exogenous variables, your unknowns, and your targets? What is a good DAG?
3. Parameterize the steady state by normalizing Y to 1 and picking reasonable values for $\lambda, r, \sigma, \phi_\pi, \phi_T, G/Y$ and B/Y (recall that with a representative agent model, $\beta(1 + r) = 1$ in steady state.)
4. Show the impulse responses of $C_t, Y_t, N_t, B_t, T_t, \pi_t, \pi_t^w, i_t$ and r_t to exogenous AR shocks to G_t, X_t and ϵ_t . Does the choice of ϕ_T matter in this model for any real variable? Why? In which direction does each shock affect each of our key observables in estimation, Y_t, π_t and i_t ?

We now estimate the model. We posit that shocks to G_t, X_t and ϵ_t follow AR(1)'s with

$$\begin{aligned} \log G_t &= \rho_G \log G_{t-1} + \eta_t^G \\ \log X_t &= \rho_X \log X_{t-1} + \eta_t^X \\ \epsilon_t &= \rho_\epsilon \epsilon_{t-1} + \eta_t^\epsilon \end{aligned}$$

where η_t^G, η_t^X and η_t^ϵ are iid normal variables with standard deviations σ_G, σ_X and σ_ϵ , respectively. We assume the following priors, as in [Herbst and Schorfheide \(2015\)](#)

Shock process parameters		Model parameters	
Param.	Prior Distribution	Param.	Prior Distribution
ρ_G	Uniform(0,1)	κ_w	Uniform(0,1)
ρ_ϵ	Uniform(0,1)	ϕ_π	Gamma(1.5,0.25)
ρ_X	Uniform(0,1)	ϕ_T	Uniform(0,1)
σ_G	InvGamma(1,4)		
σ_ϵ	InvGamma(0.4,4)		
σ_X	InvGamma(0.5,4)		

5. To test your estimation procedure, simulate data from your model of the length of the observed time series. [Hint: you can use the function `simulate` provided in the `estimation_utils.py` file for this purpose.] Then, estimate parameters of the shock processes by maximizing the likelihood. [Hint: you can use the function `loglik_f` for this purpose. This function calculates the log-likelihood and combines it with pre-specified priors to calculate a posterior. Here, for this test, we just use maximum likelihood, or equivalently uniform priors.] Do this for at least 10 different simulations. How close does the distribution of parameters that you get from this Monte Carlo procedure get to your calibrated parameters?
6. Now estimate parameters of the shock process by maximizing the posterior mode on the actual U.S. time series, using the priors in the table above. Report your estimated parameters.
7. Now estimate parameters of the shock process as well model parameters (κ_w, ϕ_π) by maximizing the posterior mode. Why can't you estimate ϕ_T ?
8. **Post-estimation steps.**
 - i. Report the impulse responses at the posterior mode. These will be useful for interpretation of the two items below.
 - ii. Perform a historical decomposition of the observed time series into the three shocks (since we have three shocks and three time series, we can do this exactly). Interpret using your answers to question 4. [Hint: You can use the `back_out_shocks` function for this.]
 - iii. Perform a forecast error variance decomposition of Y, π, i into the three shocks at various horizons. Interpret these. Note: for this question it is helpful to use the `step1_est` function, which gives the MA representation.

1.3 Heterogeneous-agent model

We now consider the heterogeneous-agent model.

9. Repeat the steady state calibration steps from question 3 for the heterogeneous agent model. Use the standard income process used in class. Pick government liquidity B/Y to achieve

a target quarterly MPC of 0.25 in your baseline calibration. What are the iMPCs in this calibration?

10. Show the impulse responses of C_t , Y_t , N_t , B_t , T_t , π_t , π_t^w , i_t and r_t to exogenous shocks to G_t , X_t and ϵ_t . How do they compare to the representative-agent model? Now why does the choice of ϕ_T matter for real variables?
11. Estimate parameters of the shock process by maximizing the posterior mode. Report your estimated parameters.
12. Estimate parameters of the shock process and model parameters $(\kappa_w, \phi_\pi, \phi_T)$ by maximizing the posterior mode. How speed gain do you get by reusing the HA Jacobian? [Hint: to reuse your HA block jacobian, just store the jacobian as a matrix J. Then, when defining your blocks for the get_G function, use J directly instead of the hetblock.]
13. Repeat the post-estimation steps from question 8. Interpret the differences in estimation outcomes between HA and RA.

1.4 Behavioral HANK model

We now want to see if sticky information helps us fit the data. Assume that our heterogeneous agents have sticky information with respect to all their perceived inputs, as in [Auclert, Rognlie and Straub \(2020\)](#), with a Calvo updating parameter or θ .

14. By manipulating your Jacobian for the HA block as described in class, show general equilibrium impulse responses to all three shocks when there is sticky information of size θ for a calibrated θ .
15. Now estimate θ along with all other parameters. Plot impulse responses at the posterior mode, with or without sticky information. Does sticky information allow you to get a better fit to the macro data? Are the impulse responses realistic?

References

- Auclert, Adrien, Matthew Rognlie, and Ludwig Straub**, “Micro Jumps, Macro Humps: Monetary Policy and Business Cycles in an Estimated HANK Model,” Working Paper 26647, National Bureau of Economic Research, January 2020.
- Herbst, Edward P. and Frank Schorfheide**, *Bayesian Estimation of DSGE Models*, Princeton University Press, December 2015. Google-Books-ID: ZDNJCgAAQBAJ.